Hospital Management System - V1

Modern Application Development I — Project Report

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Student Details

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Project Overview

Project Title: Hospital Management System - V1

Problem Statement:

The objective was to create a web-based multi-user hospital management system. The application supports three primary roles: Admin, Doctor, and Patient. Admins can manage doctors, view appointments, and search records; Doctors can set availability, manage appointments, and record treatments; Patients can register, browse doctors, book appointments, and view their medical history.

Approach:

We adopted a modular MVC-like architecture using Flask as the backend framework. The database schema was designed in SQLite with 7 interconnected tables to support all user actions, relationships, and appointment tracking. The frontend used Jinja2, HTML, CSS, and Bootstrap 5 to create a responsive and intuitive user interface. Key features include appointment conflict prevention, treatment history tracking, and role-based access control. The solution is fully functional on a local machine as per the requirements.

Frameworks and Libraries Used

Layer - Technology Used

Backend - Python, Flask 3.0.0

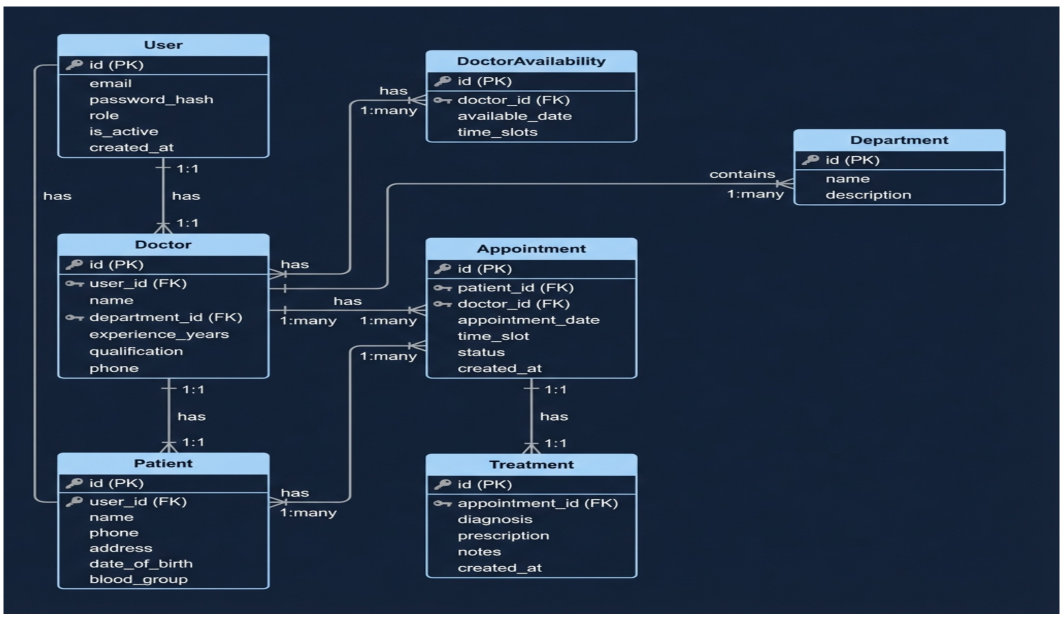
Frontend - HTML5, CSS3, Bootstrap 5, Jinja2

Database - SQLite, SQLAlchemy ORM

Security - Werkzeug (password hashing)

Additional - Bootstrap Icons

Entity-Relationship (ER) Diagram



Database Schema:

- User: Stores authentication credentials and user roles (admin/doctor/patient)

- Doctor: Doctor-specific information linked to User

- Patient: Patient-specific information linked to User

- Department: Hospital departments (Cardiology, Neurology, etc.)

- Appointment: Appointment bookings with date, time, and status

- Treatment: Medical records linked to completed appointments

- DoctorAvailability: Doctor's available time slots for next 7 days

Key Relationships:

- User → Doctor/Patient (One-to-One)

- Department → Doctor (One-to-Many)

- Doctor → Appointment (One-to-Many)

- Patient → Appointment (One-to-Many)

- Appointment → Treatment (One-to-One)

- Doctor → DoctorAvailability (One-to-Many)

Blueprint Routes

Admin Routes:

/admin/dashboard - GET - View statistics and analytics

/admin/doctors - GET - List all doctors

/admin/doctor/add - GET/POST - Add a new doctor

/admin/doctor/<id>/edit - GET/POST - Edit doctor details

/admin/doctor/<id>/delete - POST - Deactivate a doctor

/admin/appointments - GET - View all appointments

/admin/search - GET/POST - Search doctors and patients

Doctor Routes:

/doctor/dashboard - GET - View appointments and statistics

/doctor/availability - GET/POST - Set availability for next 7 days

/doctor/appointment/<id> - GET - View appointment details

/doctor/appointment/<id>/complete - POST - Add treatment and mark complete

Patient Routes:

/patient/dashboard - GET - View upcoming appointments

/patient/doctors - GET - Browse doctors by department

/patient/book/<doctor_id> - GET/POST - Book an appointment

/patient/appointments - GET - View appointment history

/patient/appointment/<id>/cancel - POST - Cancel an appointment

/patient/profile/edit - GET/POST - Edit profile information

Core Features Implemented

1. Appointment Conflict Prevention

   - System checks existing bookings before allowing new appointments
   - Prevents double-booking of same time slot for a doctor
   - Validates date and time slot availability in real-time

2. Treatment History Tracking

   - Complete medical records maintained for each patient
   - Doctors can add diagnosis, prescription, and notes
   - Patients can view their complete treatment history

3. Role-Based Access Control

   - Three distinct user roles with specific permissions
   - Session-based authentication with password hashing
   - Custom decorators to protect routes based on user role

4. Doctor Availability Management

- Doctors set availability for next 7 days
- Patients see only available time slots when booking
- Dynamic availability updates based on bookings

5. Search Functionality
   - Admin can search doctors by name or specialization
   - Admin can search patients by name, phone, or ID
   - Real-time search results display

6. UI Enhancements
   - Gradient backgrounds on dashboard stat cards
   - Hover effects (cards lift up on hover)
   - Clean spacing and modern typography
   - Responsive design for all screen sizes

Project Demo Video

Drive Link: https://drive.google.com/file/d/1WYBkdUHUQt80a-cEoKpzu_ezc2pAHLy4/view?usp=drive_link

The video covers:

- Project overview & approach
- Admin walkthrough
- Doctor walkthrough
- Patient walkthrough
- Demonstration of appointment conflict prevention
- UI design improvements

Summary

The project meets all required core functionalities and adds enhancements like gradient UI design, comprehensive treatment tracking, and robust conflict prevention. It is user-friendly, secure, and modular.

Key Achievements:

- 7-table relational database
- Full CRUD operations
- Secure authentication
- Appointment conflict prevention
- Treatment history tracking
- Responsive UI with Bootstrap 5
- Role-based access control